

Q.1 Consider the following C program. Assume parameters to a function are evaluated from right to left. **(GATE CSE 2024 Set 2)**

```
#include <stdio.h>
int g(int p) {
    printf("%d", p); return p;
}
int h(int q) {
    printf("%d", q); return q;
}

void f(int x, int y) {
    g(x);
    h(y);
}

int main() {
    f(g(10),h(20));
}
```

Which one of the following options is the CORRECT output of the above C program?

- (A) 20101020
- (B) 10202010
- (C) 20102010
- (D) 10201020

Answer : (A) 20101020

Q.2 Consider the following C program: **(GATE CSE 2024 Set 1)**

```
#include <stdio.h>
void fX();
int main() {
    fX();
    return 0;
}

void fX() {
    char a;
    if ((a = getchar()) != '\n')
        fX();
    if (a != '\n')
```

```
    putchar(a);
}
```

Assume that the input to the program from the command line is 1234 followed by a newline character. Which one of the following statements is CORRECT?

- (A) The program will not terminate
- (B) The program will terminate with no output
- (C) The program will terminate with 4321 as output
- (D) The program will terminate with 1234 as output

Answer : (C) The program will terminate with 4321 as output

Q.3 The integer value printed by the ANSI-C program given below is _____.

(GATE CSE 2023)

```
#include<stdio.h>
int funcp(){
    static int x = 1;
    x++ ;
    return x;
}
int main(){
    int x,y;
    x = funcp();
    y = funcp()+x;
    printf("%d\n", (x+y));
    return 0;
}
```

Answer: 7

Q.4 Consider the following ANSI C function:

(GATE CSE 2021 Set 2)

```
int SomeFunction (int x, int y)
{
    if ( (x == 1) || (y == 1))
        return 1;
    if (x == y)
        return x;
    if (x > y)
        return SomeFunction(x - y, y);
    if (y > x)
        return SomeFunction(x, y - x);
}
```

The value returned by SomeFunction(15, 255) is _____.

Answer: 15

Q.5 Consider the following C program:

(GATE CSE 2019)

```
#include <stdio.h>
int jumble (int x, int y) {
    x = 2 * x + y;
    return x;
}
int main () {
    int x=2, y=5;
    y = jumble (y, x);
    x = jumble (y, x);
    printf ("%d \n", x);
    return 0;
}
```

The value printed by the program is _____.

Answer: 26

Q.6 Consider the following C program:

(GATE CSE 2018)

```
#include <stdio.h>
int counter = 0;
int calc (int a, int b) {
    int c;
    counter++;
    if (b==3) return (a*a*a);
    else {
        c = calc(a, b/3);
        return (c*c*c);
    }
}
int main (){
    calc(4, 81);
    printf ("%d", counter);
}
```

The output of this program is _____.

Answer : 4

Q.7 Consider the following C program segment.

(GATE CSE 2015 Set 3)

```
#include <stdio.h>
int main()
{
    char s1[7] = "1234", *p;
    p = s1 + 2;
    *p = '0';
    printf("%s", s1);
}
```

What will be printed by the program?

- (A) 12
- (B) 120400
- (C) 1204
- (D) 1034

Answer: (C) 1204

Q.8 Consider the following function written in the C programming language.

.

(GATE CSE 2015 Set 2)

```
void foo(char *a){
    if ( *a && *a != ' '){
        foo(a+1);
        putchar(*a);
    }
}
```

The output of the above function on input "ABCD EFGH" is

- (A) ABCD EFGH
- (B) ABCD
- (C) HGFE DCBA
- (D) DCBA

Answer: (D) DCBA

Q.9 What will be the output of the following C program segment?

(GATE CSE 2012)

```
Char inchar = 'A';
Switch ( inchar ) {
    case 'A' : printf ("Choice A\ n") ;
    case 'B' :
    case 'C' : printf ("Choice B") ;
```

```

case 'D' :
case 'E' :
default : printf ( " No Choice" ); }

```

(A) No Choice
 (B) Choice A
 (C) Choice A
 Choice B No Choice
 (D) Program gives no output as it is erroneous

Answer: (C) Choice A

Choice B No Choice

Q.10 Which one of the following are essential features of an object-oriented programming language? **(GATE CSE 2005)**

- i) Abstraction and encapsulation
 - ii) Strictly-typedness
 - iii) Type-safe property coupled with sub-type rule
 - iv) Polymorphism in the presence of inheritance
- (A) (i) and (ii) only
 (B) (i) and (iv) only
 (C) (i), (ii) and (iv) only
 (D) (i), (ii) and (iii) only

Answer: (B) (i) and (iv) only

Q.11 An Abstract Data Type (ADT) is **(GATE CSE 2005)**

- (A) same as an abstract class
 (B) a data type that cannot be instantiated
 (C) a data type for which only the operations defined on it can be used, but none else
 (D) all of the above

Answer : (C) a data type for which only the operations defined on it can be used, but none else

Q.12 Consider the following C function. **(GATE CSE 2003)**

```

float f,(float x, int y) {
    float p, s; int i;
    for (s=1,p=1,i=1; i < y; i++) {
        p *= x/i;
        s+=p;
    }
}

```

```
return s;  
}
```

For large values of y , the return value of the function f best approximates

- (A) X^Y
- (B) e^x
- (C) $\ln(1+x)$
- (D) X^X

Answer : (B) e^x

Q.13 In the C language

(GATE CSE 2002)

- (A) At most one activation record exists between the current activation record and the activation record for the main
- (B) The number of activation records between the current activation record and the activation record of the main depends on the actual function calling sequence.
- (C) The visibility of global variables depends on the actual function calling sequence.
- (D) Recursion requires the activation record for the recursive function to be saved on a different stack before the recursive fraction can be called.

Answer: (B) The number of activation records between the current activation record and the activation record of the main depends on the actual function calling sequence.

Q.14 Consider the following ANSI-C function.

(GATE CSE 2026 Set 2)

```
int func(int start, int end){  
  
    int length = end + 1 - start;  
  
    if((length<1) || (start < 0) || (end<0)){  
        return(0);  
    }  
    if(length%3 == 0){  
        return(func(start+1, end));  
    }  
    else if(length%3 == 1){  
        return(1 + func(start, end -1));  
    }  
    else {  
        return(func(start + 2, end));  
    }  
}
```

```
}  
}
```

The maximum possible value that can be returned from this function is . (answer in integer)

Answer: 1

Q.15 Consider the recursive functions represented by the following code segment:

(GATE CSE 2026 Set 1)

```
int bar(int n){  
  
    if (n == 1)  
        return 0;  
  
    else  
        return 1 + bar(n/2);  
  
}
```

```
int foo(int n){  
  
    if (n == 1)  
        return 1;  
  
    else  
        return 1 + foo(bar(n));  
  
}
```

The smallest positive integer n for which foo(n) returns 5 is . (answer in integer)

Answer: 65536

Q.16 Consider the following C function definition.

(GATE CSE 2024 Set 1)

```
int f(int x, int y) {  
    for (int i=0; i<y; i++) {  
        x=x+x+y;  
    }  
    return x;  
}
```

Which of the following statements is/are TRUE about the above function?

(A) If the inputs are $x = 20$ $y = 10$ then the return value is greater than 220

(B) If the inputs are $x = 20$ $y = 20$ then the return value is greater than 220

(C) If the inputs are $x = 20$ $y = 10$ then the return value is less than 210

(D) If the inputs are $x = 10$ $y = 20$ then the return value is greater than 220

Answer: (B) If the inputs are $x = 20$ $y = 20$ then the return value is greater than 220

(D) If the inputs are $x = 10$ $y = 20$ then the return value is greater than 220

Q.17 What is printed by the following ANSI C program?

(GATE CSE 2022)

```
#include <stdio.h>
int main(int argc, char *argv[ ]) {
    int a[3][3][3] = {{1, 2, 3, 4, 5, 6, 7, 8, 9},
        {10, 11, 12, 13, 14, 15, 16, 17, 18},
        {19, 20, 21, 22, 23, 24, 25, 26, 27}};
    int i = 0, j = 0, k = 0;
    for(i = 0; i < 3; i++) {
        for(k = 0; k < 3; k++)
            printf("%d", a[i][j][k]);
        printf("\n");
    }
    return 0;
}
```

Answer: 1 2 3

10 11 12

19 20 21

Q.18 Consider the following ANSI C program

(GATE CSE 2021 Set 2)

```
#include <stdio.h>
int foo(int x, int y, int q)
{
    if ((x <= 0 && (y <= 0))
        return q;
    if (x <= 0)
        return foo(x, y - q, q);
    if (y <= 0)
        return foo(x - q, y, q);
    return foo(x - q, y - q, q) + foo(x - q, y, q);
}
int main()
(
    int r = foo(15, 15, 10);
    printf("%d", r);
}
```



```
return 0;
}
```

The output of the program upon execution is _____

Answer: 60

Q.19 Consider the following ANSI C program.

(GATE CSE 2021 Set 1)

```
#include <stdio.h>
int main( ) {
    int i, j, count;
    count = 0;
    i = 0;
    for (j = -3; j <= 3; j++) {
        if ((j >= 0) && (i++))
            count = count + j;
    }
    count = count + i;
    printf("%d", count);
    return 0;
}
```

Which one of the following options is correct?

- (A) The program will compile successfully and output 10 when executed.
- (B) The program will compile successfully and output 8 when executed.
- (C) The program will compile successfully and output 13 when executed.
- (D) The program will not compile successfully.

Answer: (A) The program will compile successfully and output 10 when executed.

Q.20 Consider the following ANSI C function:

(GATE CSE 2021 Set 1)

```
int SimpleFunction (int y[], int n, int x)
{
    int total = y[0], loopIndex;
    for (loopIndex = 1; loopIndex <= n - 1; loopIndex++)
        total = x * total + y[loopIndex];
    return total ;
}
```

Let Z be an array of 10 elements with $Z[i] = 1$, for all i such that $0 \leq i \leq 9$. The value returned by SimpleFunction (Z, 10, 2) is _____

Answer: 1023

Q.21 Consider the following C functions.

(GATE CSE 2020)

```
int tob(int b, int* arr) {
    int i;
    for (i=0; b>0; i++) {
        if (b%2) arr[i] = 1;
        else arr[i]=0;
        b = b/2;
    }
    return (i);
}
```

```
int pp(int a, int b) {
    int arr[20];
    int i, tot = 1, ex, len;
    ex = a;
    len = tob (b, arr);
    for (i=0; i < len; i++) {
        if (arr[i] ==1)
            tot = tot * ex;
        ex = ex * ex;
    }
    return (tot);
}
```

The value returned by pp (3, 4) is _____.

Answer: 81

Q.22 Consider the following C functions.

(GATE CSE 2020)

```
int fun1 (int n) {
    static int i = 0;
    if (n > 0) {
        ++i;
        fun1(n-1);
    }
    return (i);
}
```

```
int fun2 (int n) {
    static int i = 0;
```

```

if (n > 0) {
    i = i + fun1 (n);
    fun2(n-1);
}
return (i);
}

```

The return value of fun2 (5) is _____.

Answer: 55

Q.23 Consider the following C function.

(GATE CSE 2019)

```

void convert(int n){
    if(n < 0)
        printf("%d",n);
    else {
        convert(n/2);
        printf("%d",n%2);
    }
}

```

Which one of the following will happen when the function convert is called with any positive integer n as argument?

- (A) It will print the binary representation of n and terminate
- (B) It will print the binary representation of n in the reverse order and terminate
- (C) It will print the binary representation of n but will not terminate
- (D) It will not print anything and will not terminate

Answer: (D) It will not print anything and will not terminate

Q.24 Consider the following C program :

(GATE CSE 2019)

```

#include<stdio.h>
int r()
{
    static int num=7;
    return num--;
}

int main()
{

```

```

    for(r();r();r())
    printf("%od ",r());
    return 0;
}

```

Which one of the following values will be displayed on execution of the programs?

- (A) 41
- (B) 52
- (C) 63
- (D) 630

Answer: (B) 52

Q.25 Consider the following C code. Assume that *unsigned long int* type length is 64 bits. (GATE CSE 2018)

```

unsigned long int fun(unsigned long int n){
    unsigned long int i, j = 0, sum = 0;
    for (i = n; i > 1; i = i/2) j++;
    for ( ; j > 1; j = j/2) sum++;
    return(sum);
}

```

The value returned when we call *fun* with the input 2^{40} is

- (A) 4
- (B) 5
- (C) 6
- (D) 40

Answer: (B) 5

Q.26 Consider the C program below. (GATE CSE 2015 Set 2)

```

#include < stdio.h >
int *A, stkTop;
int stkFunc(int opcode, int val)
{
    static int size=0, stkTop=0;
    switch (opcode) {
        case -1: size = val; break;
        case 0: if (stkTop < size) A[stkTop++] = val; break;
        default: if (stkTop) return A[--stkTop];
    }
    return -1;
}

```

```

int main()
{
    int B[20]; A = B; stkTop = -1;
    stkFunc (-1, 10);
    stkFunc ( 0, 5);
    stkFunc ( 0, 10);
    printf ("%d\n", stkFunc(1, 0) + stkFunc(1, 0));
}

```

The value printed by the above program is _____.

Answer: 15

Q.27 Consider the program given below, in a block-structured pseudo-language with lexical scoping and nesting of procedures permitted. **(GATE CSE 2012)**

Program main;

Var . . .

Procedure A1;

Var . . .

Call A2;

End A1

Procedure A2;

Var . . .

Procedure A21;

Var . . .

Call A1;

End A21

Call A21;

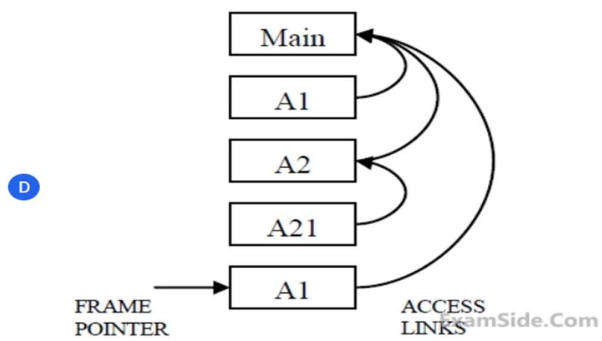
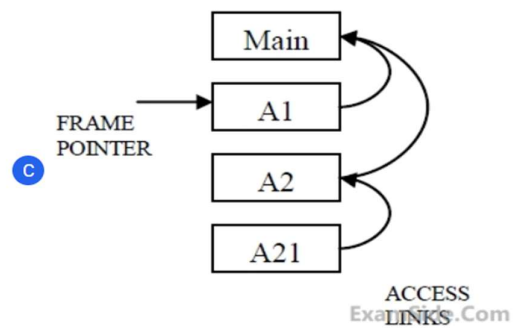
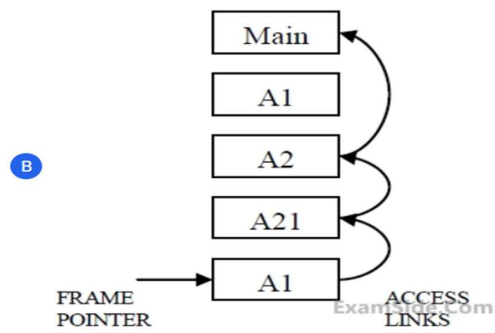
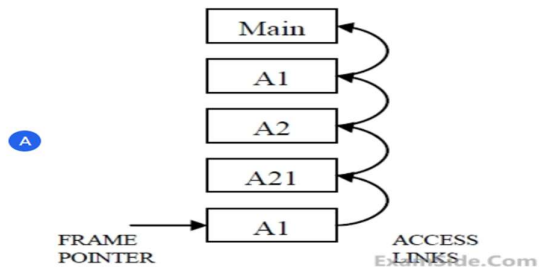
End A2

Call A1;

End main.

Consider the calling chain: Main -> A1 -> A2 -> A21 -> A1

The correct set of activation records along with their access links is given by



Answer: (D)

Q.28 Consider the following recursive C function that takes two arguments:

(GATE CSE 2011)

```
unsigned int foo (unsigned int n, unsigned int r) {  
    if (n > 0) return((n % r) + foo(n/r, r));  
    else return 0;  
}
```

What is the return value of the function foo when it is called as foo (513, 2)?

- (A) 9
- (B) 8
- (C) 5
- (D) 2

Answer: 2

Q.29 Consider the following recursive C function that takes two arguments:

(GATE CSE 2011)

```
unsigned int foo (unsigned int n, unsigned int r) {  
    if (n > 0) return((n % r) + foo(n/r, r));  
    else return 0;  
}
```

What is the return value of the function foo when it is called as foo (345, 10)?

- (A) 345
- (B) 12
- (C) 5
- (D) 3

Answer: (B) 12

Q.30 Choose the correct option to fill ? 1 and ? 2 so that the program below prints an input string in reverse order. Assume that the input string is terminated by a newline character.

(GATE CSE 2008)

```
void reverse (void) {  
    int c;  
    if (?1) reverse();  
    ?2  
}  
main {  
    printf ("Enter Text" );  
    printf("\n");  
    reverse();
```

```
printf("\n");  
}
```

- (A) ?1 is (getchar() != '\n')
 ?2 is getchar (c);
- (B) ?1 is ((c = getchar()) != '\n')
 ?2 is getchar (c);
- (C) ?1 is (c != '\n')
 ?2 is putchar (c);
- (D) ?1 is ((c = getchar()) != '\n')
 ?2 is putchar (c);

Answer: (D) ?1 is ((c = getchar()) != '\n')
?2 is putchar (c);

Q.31 Consider the following C function

(GATE CSE 2007)

```
int f(int n)  
{  
    static int r = 0;  
    if(n <= 0) return 1;  
    if(n>3)  
    {  
        r = n;  
        return f(n-2)+2;  
    }  
    return f(n-1)+r;  
}
```

What is the value of f(5)?

- (A) 5
- (B) 7
- (C) 9
- (D) 18

Answer: (D) 18

Q.32 Consider the following C program:

(GATE CSE 2005)

```
void foo (int n, int sum) {  
    int k = 0, j = 0;  
    if(n==0) return;  
    k=n%10; j = n /10;  
    sum = sum + k;  
    foo(j, sum);  
}
```



```

    printf("%d",k);
}
int main () {
    int a = 2048, sum = 0;
    foo(a, sum);
    printf("%d\n", sum);
}

```

What does the above program print?

- (A) 8, 4, 0, 2, 14
- (B) 8, 4, 0, 2, 0
- (C) 2, 0, 4, 8, 14
- (D) 2, 0, 4, 8, 0

Answer: (D) 2, 0, 4, 8, 0

Q.33 Consider the following C program:

(GATE CSE 2005)

```

double foo(double); /* Line 1 */
int main () {
    double da, db;
    //input da
    db = foo(da);
}
double foo(double a){
    return a;
}

```

The above code compiled without any error or warning. If Line 1 is deleted, the above code will show:

- (A) No compile warning or error
- (B) Some compiler-warnings not leading to unintended results
- (C) Some compiler-warnings due to type-mismatch eventually leading to unintended results
- (D) Compiler errors

Answer: (D) Compiler errors

Q.34 Consider the following C function

(GATE CSE 2004)

```

int f(int n)
{
    static int i = 1;
    if(n>=5) return n;
    n = n+1;
    i++;
}

```

```
return f(n);  
}
```

The value returned by f(1) is

- (A) 5
- (B) 6
- (C) 7
- (D) 8

Answers: (A) 5

Q.35 Consider the following C program

(GATE CSE 2004)

```
main ( )  
{  
    int x, y, m, n;  
    scanf("%d %d", &x, &y);  
    /* Assume x > 0 and y > 0 */  
    m=x; n=y;  
    while(m!=n)  
    {  
        if(m>n)  
            m=m-n;  
        else  
            n=n-m;  
    }  
    printf("%d", n);  
}
```

The program computes

- (A) $x + y$ using repeated subtraction
- (B) $x \bmod y$ using repeated subtraction
- (C) the greatest common divisor of x and y
- (D) the least common multiple of x and y

Answer: (C) the greatest common divisor of x and y

Q.36 Consider the following class definitions in a hypothetical Object Oriented language that supports inheritance and uses dynamic binding. The language should not be assumed to be either Java or C++, though the syntax is similar. **(GATE CSE 2003)**

```
Class P {  
    void f(int i) {  
        print(i);  
    }  
}
```

```
}
```

```
Class Q subclass of P {
```

```
    void f(int i) {
```

```
        print(2*i);
```

```
    }
```

```
}
```

Now consider the following program fragment:

```
Px = new Q();
```

```
Qy = new Q();
```

```
Pz = new Q();
```

```
x.f(1); ((P)y).f(1); z.f(1);
```

Here ((P)y) denotes a typecast of y to P. The output produced by executing the above program fragment will be

(A) 1 2 1

(B) 2 1 1

(C) 2 1 2

(D) 2 2 2

Answer: (C) 2 1 2

Q.37 Consider the following declaration of a two-dimensional array in C:

```
char a[100][100];
```

Assuming that the main memory is byte-addressable and that the array is stored starting from memory address 0, the address of a[40][50] is **(GATE CSE 2002)**

(A) 4040

(B) 4050

(C) 5040

(D) 5050

Answer: (B) 4050

Q.38 The value of j at the end of the execution of the following C program is

(GATE CSE 2000)

```
int incr (int i)
```

```
{
```

```
    static int count = 0;
```

```
    count = count + i;
```

```
    return (count);
```

```
}
```

```
main () {
    int i,j;
    for (i = 0; i <= 4; i++)
        j = incr(i);
}
```

- (A) 10
- (B) 4
- (C) 6
- (D) 7

Answer: (A) 10

Q.39 Consider the following C function definition

(GATE CSE 1999)

```
int Trial (int a, int b, int c)
{
    if ((a >= b) && (c < b)) return b;
    else if (a >= b) return Trial (a,c,b);
    else return Trial (b,a,c);
}
```

The function Trial:

- (A) Finds the maximum of a, b and c
- (B) Finds the minimum of a, b and c
- (C) Finds the middle number of a, b and c
- (D) None of the above

Answer: (C) Finds the middle number of a, b and c

Q.40 Let P be an array containing n integers. Let t be the lowest upper bound on the number of comparisons of the array elements, required to find the minimum and maximum values in an arbitrary of n elements. Which one of the following choices is correct?

(GATE CSE 2021 Set 1)

- (A) $t > 2n-2$
- (B) $t > 3\lceil n/2 \rceil$ and $t \leq 2n-2$
- (C) $t > n$ and $t \leq 3\lceil n/2 \rceil$
- (D) $t > \lceil \log_2(n) \rceil$ and $t \leq n$

Answer: (C) $t > n$ and $t \leq 3\lceil n/2 \rceil$

Q.41 Consider the following C program.

(GATE CSE 2020)

```
#include <stdio.h>
```

```

int main () {
    int a [4] [5] = {{1, 2, 3, 4, 5},
                     {6, 7, 8, 9, 10},
                     {11, 12, 13, 14, 15},
                     {16, 17, 18, 19, 20}};
    printf ("%d\n", *(a+**a+2) +3));
    return (0);
}

```

The output of the program is _____.

Answer: 19

Q.42 A program P reads in 500 integers in the range [0, 100] representing the cores of 500 students. It then print the frequency of each score above 50. What would be the best way for P to store the frequencies? **(GATE CSE 2005)**

- (A) An array of 50 numbers
- (B) An array of 100 numbers
- (C) An array of 500 numbers
- (D) A dynamically allocated array of 550 numbers

Answer: (A) An array of 50 numbers

Q.43 A single array A[1..MAXSIZE] is used to implement two stacks, The two stacks grow from opposite ends of the array. Variables top1 and top2 (top1 < top2) point to the location of the topmost element in each of the stacks, If the space is to be used efficiently, the condition for "stack full" is **(GATE CSE 2004)**

- (A) (top 1 = MAXSIZE/2) and (top 2 = MAXSIZE/2 + 1)
- (B) top 1 + top 2 = MAXSIZE
- (C) (top 1 = MAXSIZE/2) or (top 2 = MAXSIZE)
- (D) top 1 = top 2 - 1

Answer: (D) top 1 = top 2 – 1

Q.44 Suppose you are given an array s[1..n] and a procedure reverse (s, i, j) which reverse the order of elements in s between positions i and j (both inclusive). What does the following sequence do, where reverse (s, 1, k);

reverse (s, k+1, n);

reverse (s, 1, n);

(GATE CSE 2000)

- (A) Rotates s left by k positions
- (B) Leaves s unchanged
- (C) Reverse all elements of s
- (D) None of the above

Answer: (A) Rotates s left by k positions

Q.45 An $n \times n$ array v is defined as follows $V[i, j] = i - j$ for all i, j . The sum of the elements of the array v is **(GATE CSE 2000)**

- (A) 0
- (B) $n-1$
- (C) $n^2 - 3n + 2$
- (D) $n^2(n+1)/2$

Answer: (A) 0

Q.46 An array is heapified. Which one of the following options represents the first three elements in the heapified array? **(GATE CSE 2024 Set 1)**

- (A) 82 90 101
- (B) 82 11 93
- (C) 131 11 93
- (D) 131 111 90

Answers: (D) 131 111 90

Q.47 Consider the following C functions in which size is the number of elements in the array E **(GATE CSE 2014 Set 1)**

```
int MyX(int *E, unsigned int size){
    int Y = 0;
    int Z;
    int i,j,k;
    for(i = 0; i < size; i++)
        Y = Y + E[i];
    for(i = 0; i < size; i++)
        for(j = i; j < size; j++){
            Z = 0;
            for(k = i; k <= j; k++)
                Z = Z + E[k];
            if(Z > Y)
                Y = Z;
        }
    return Y;
}
```

The value returned by the function MyX is the

- (A) Maximum possible sum of elements in any sub-array of array E
- (B) Maximum elements in any sub-array of array E
- (C) Sum of the maximum elements in all possible sub-arrays of array E

(D) The sum of all elements in the array E

Answer: (A) Maximum possible sum of elements in any sub-array of array E

Q.48 Two matrices M_1 and M_2 are to be stored in arrays A and B respectively. Each array can be stored either in row-major or column-major order in contiguous memory locations. The time complexity of an algorithm to compute $M_1 M_2$ will be

(GATE CSE 2004)

- (A) Best if A is in row-major and B is in column-major order
- (B) Best if both are in row-major order
- (C) Best if both are in column-major order
- (D) Independent of the storage scheme

Answer: (D) Independent of the storage scheme

Q.49 Let A be a two dimensional array declared as follows:

A : array [1... 10] [1... 15] of integer;

Assuming that each integer takes one memory locations the array is stored in row-major order and the first element of the array is stored at location 100, what is the address of the element A[i] [j]?

(GATE CSE 1998)

- (A) $15i + j + 84$
- (B) $15j + i + 84$
- (C) $10i + j + 89$
- (D) $10j + i + 89$

Answer: (A) $15i + j + 84$

Q.50 In a compact single dimensional array representation for lower triangular matrices (i.e. all the elements above the diagonal are zero) of size $n \times n$, non-zero elements (i.e., elements of the lower triangle) of each row are stored one after another, starting from the first row, the index of the $(i, j)^{\text{th}}$ element of the lower triangular matrix in this new representation is

(GATE CSE 1994)

- (A) $i + j$
- (B) $i + j - 1$
- (C) $(j - 1) + i(i - 1)/2$
- (D) $i + j(j - 1)/2$

Answer: (C) $(j - 1) + i(i - 1)/2$